Experiment No.	1
Experiment Title	Working with Python Packages– NumPy, SciPy, Scikit Learn, Matplotlib (Exploratory Data Analysis and Prepro cessing Pipeline) [1]
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1 Objective

To explore and understand the core Python libraries used in machine learning workflows – NumPy, SciPy, Pandas, Matplotlib, Seaborn, and Scikit-Learn – by building a generic, reusable Exploratory Data Analysis (EDA) and preprocessing pipeline, and applying it across five diverse datasets (Iris, Loan Prediction, Pima Indians Diabetes, Email Spam, and MNIST) to observe how the same pipeline behaves on structured, high-dimensional, and mixed-type data.

2 Problem Statement

Given five raw CSV datasets of varying size, dimensionality, and target type, the goal is to design a single automated pipeline that:

- Loads each dataset and reports its shape, data types, and summary statistics.
- Identifies missing values and duplicate rows.
- Automatically detects whether the underlying task is classification or regression based on the target column's data type and cardinality.
- Handles missing values (including domain-specific cases such as biologically impossible zeros in the Diabetes dataset) and caps outliers using the IQR method.
- Scales numeric features using standardization.
- Produces visualizations (histograms, correlation heatmaps, pair plots) to support interpretation.

Input: raw CSV files. Output: cleaned, scaled DataFrames along with the detected task type and diagnostic plots for each dataset.

3 Dataset Description

[1] **GitHub Repository:** https://github.com/preethi340/Machine-Learning-.git

Dataset	Description
Iris	150 samples, 6 columns (Id + 4 numeric flower measurements + Species). Target: `Species` (3 classes). No missing values.
Loan Prediction	614 samples, 13 columns of applicant and loan details (income, credit history, property area, etc.). Target used: `LoanAmount` (regression, 203 unique values). Contains missing values in several numeric and categorical columns.
Pima Indians Diabetes	768 samples, 9 columns of medical measurements. Target: `Outcome` (binary classification). No literal NaNs, but biologically invalid zeros present in Glucose, BloodPressure, SkinThickness, Insulin, and BMI.
Email Spam	5,172 samples, 3,002 columns (word-frequency features + `Email No.` + `Prediction`). Target: `Prediction` (binary classification, spam vs. ham). No missing values.
MNIST (sampled)	2,000-row sample (out of the full dataset) with 785 columns (784 pixel intensity features + `label`). Target: `label` (10-class digit classification, 0–9). Many pixel columns are constant (all zero) in the sample.

Dataset Source: UCI Machine Learning Repository / Kaggle (standard benchmark datasets).
Train-Test Split: Not performed in this experiment – the focus is on EDA and preprocessing only; splitting is deferred to later modelling experiments.

4 Exploratory Data Analysis

A single reusable function `eda_process()` was written and applied to all five datasets. For each dataset it printed the shape, first few rows, column data types, `describe()` summary statistics, missing-value counts, and duplicate-row count, and then generated histograms, a correlation heatmap, and (where the number of numeric features was small enough) a pair plot.

Iris Dataset

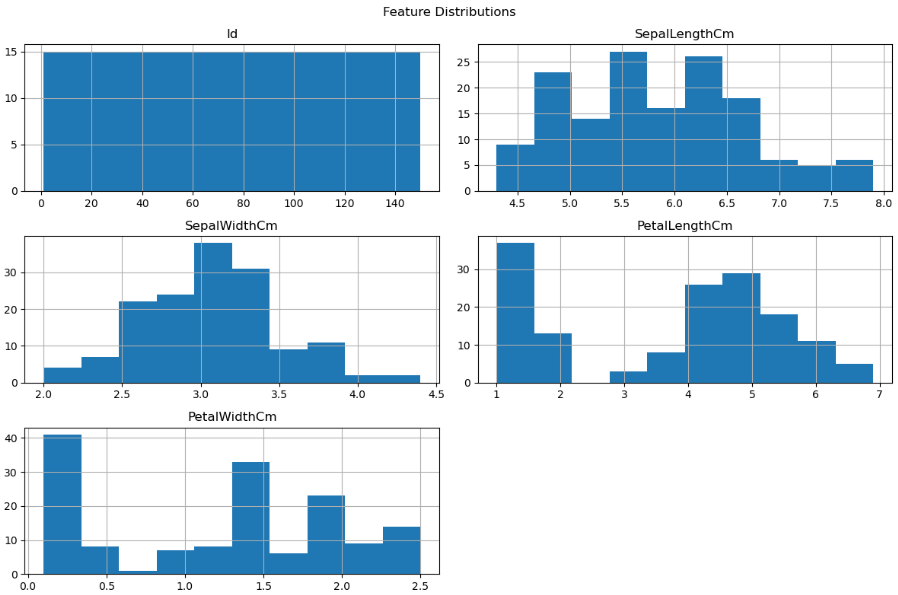

Figure 1: Feature distributions for the Iris dataset

Inference: Sepal length and width follow a roughly bell-shaped, near-normal distribution, while petal length and petal width are clearly bimodal/trimodal, reflecting the separation between *Iris-setosa* and the other two species. This early visual cue already hints that petal measurements will be far more useful for classification than sepal measurements.

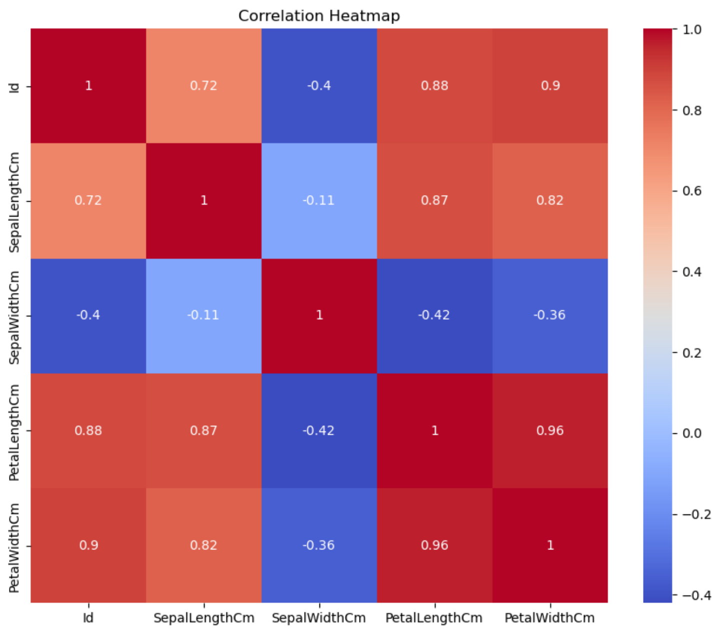

Figure 2: Correlation heatmap of Iris numeric features

Inference: Petal length and petal width are very strongly positively correlated (correlation close to 0.96), and both are strongly correlated with sepal length. Sepal width, on the other hand, is weakly and even slightly negatively correlated with the other features. This suggests some redundancy between the petal measurements.

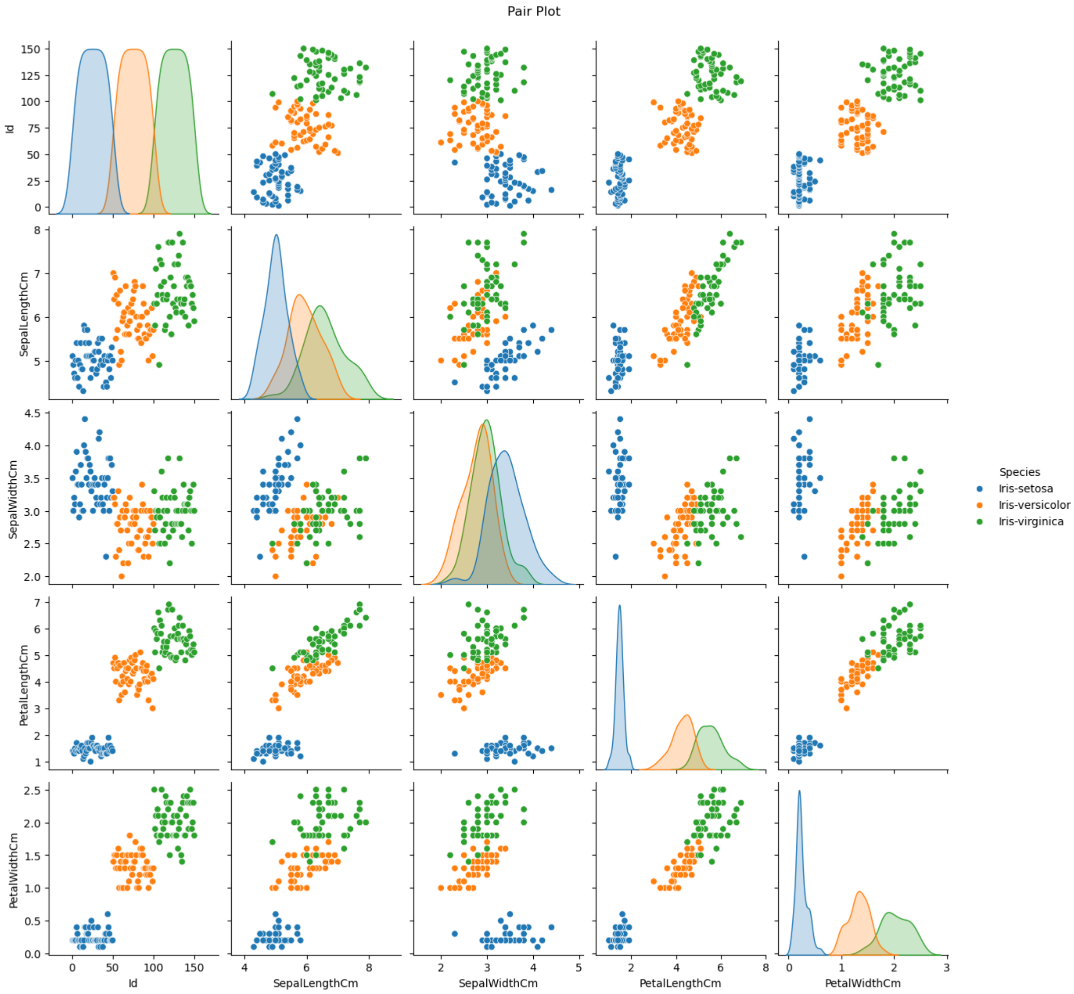

Figure 3: Pair plot of Iris features coloured by Species

Inference: The pair plot shows that *Iris-setosa* is linearly separable from the other two species on almost every feature pair, while *Iris-versicolor* and *Iris-virginica* overlap partially, especially on sepal measurements. Petal length vs. petal width gives the cleanest visual separation of all three classes.

Loan Prediction Dataset

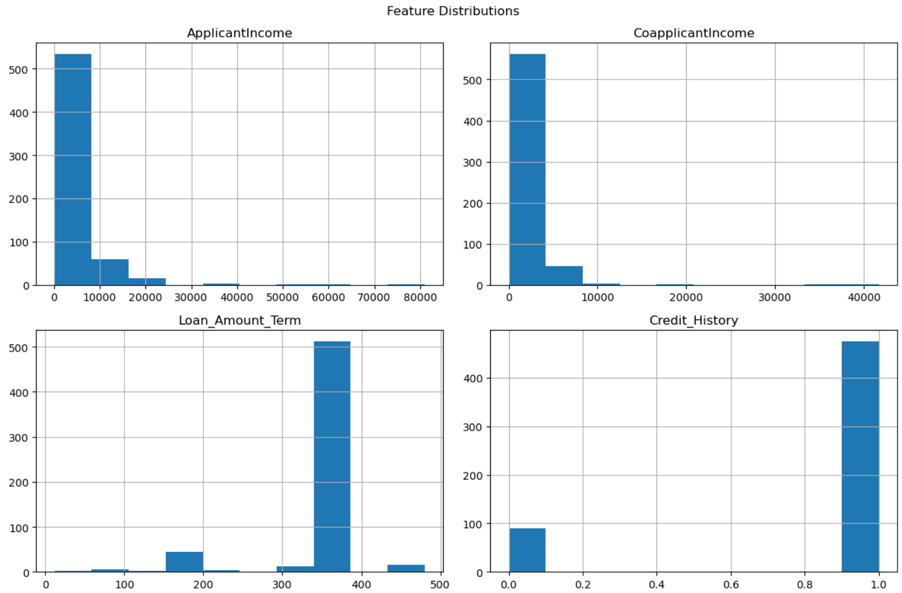

Figure 4: Feature distributions for the Loan Prediction dataset

Inference: ApplicantIncome and CoapplicantIncome are heavily right-skewed, with most applicants earning modest incomes and a small number of high earners forming a long tail. Loan_Amount_Term is highly concentrated around 360 months, and Credit_History is almost binary in practice, with most applicants having a value of 1.

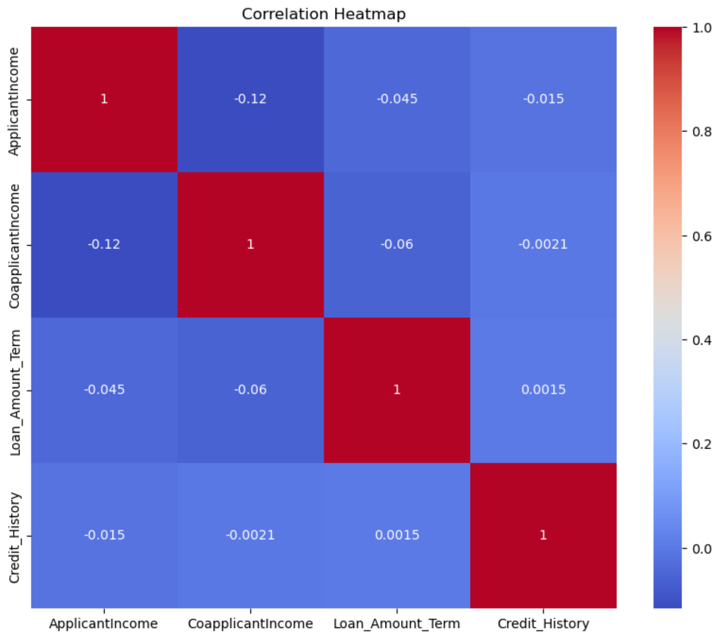

Figure 5: Correlation heatmap of Loan Prediction numeric features

Inference: None of the numeric predictors show a strong linear correlation with each other, which suggests limited multicollinearity in this subset of features. This also implies that a simple linear model based only on these numeric columns may struggle to explain

LoanAmount well, since the categorical columns (Education, Property_Area, etc.) likely carry more signal.

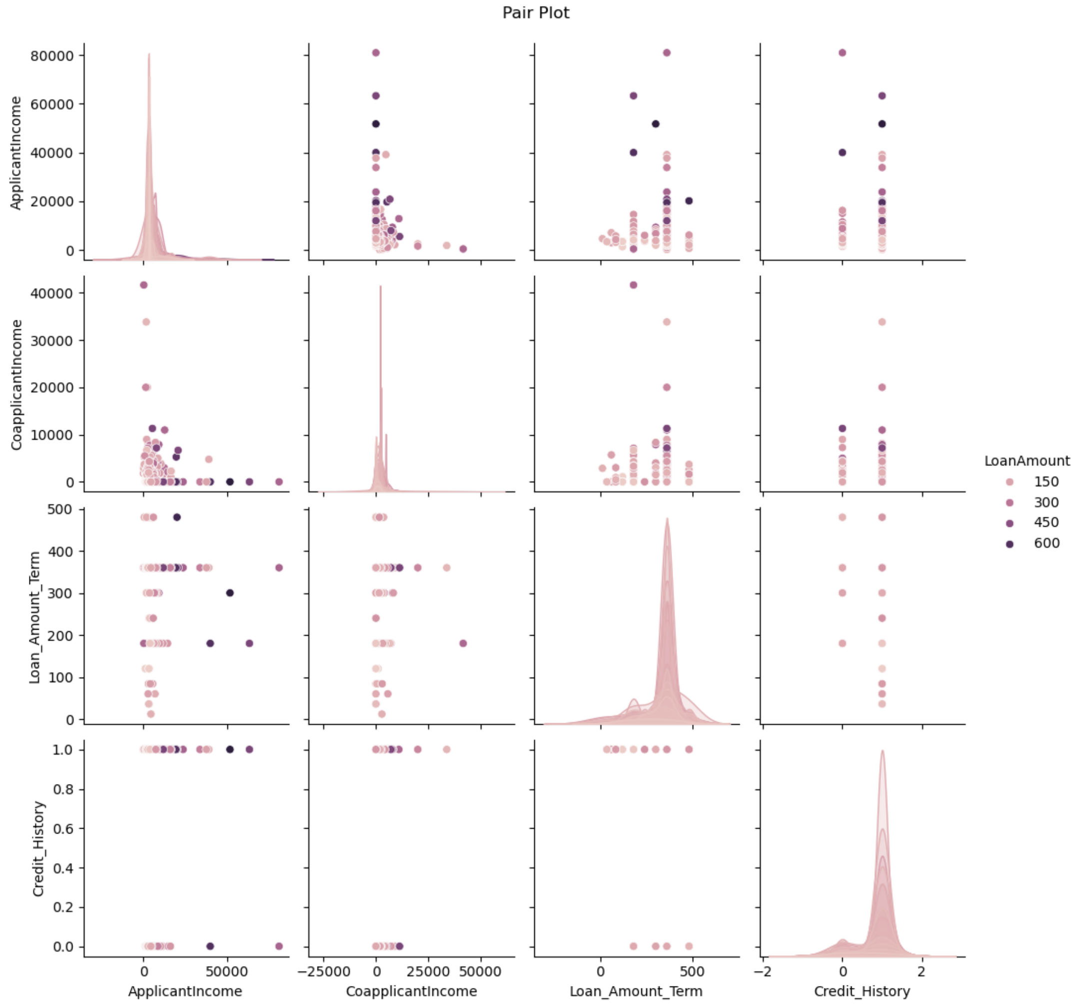

Figure 6: Pair plot of Loan Prediction numeric features

Inference: The pair plot does not reveal any clean clusters or linear trends among ApplicantIncome, CoapplicantIncome, Loan_Amount_Term, and Credit_History, reinforcing that this dataset's numeric features alone are only weakly predictive and that categorical/engineered features would be needed for a strong model.

Diabetes Dataset

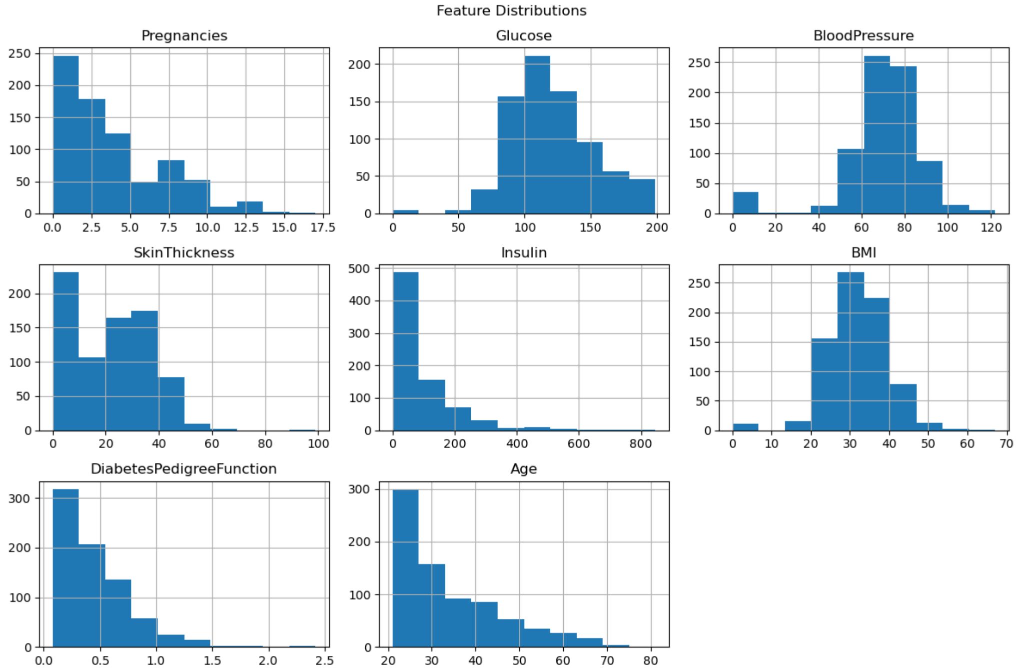

Figure 7: Feature distributions for the Diabetes dataset

Inference: Several clinically implausible values are visible as spikes at zero for Glucose, BloodPressure, SkinThickness, Insulin, and BMI – these are missing values encoded as zero rather than true measurements. Age and Pregnancies are right-skewed, and DiabetesPedigreeFunction is also skewed with a long tail, indicating a need for outlier handling before modelling.

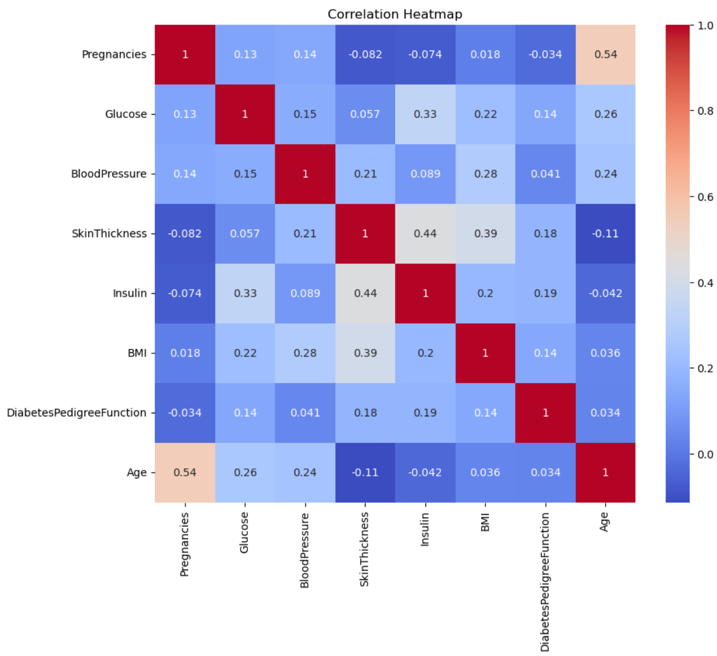

Figure 8: Correlation heatmap of Diabetes dataset features

Inference: Glucose shows the strongest correlation with the Outcome (diabetes) label among all features, consistent with its clinical role in diagnosis. BMI and Age also show moderate positive correlations with Outcome, while SkinThickness and Insulin are strongly correlated with each other, suggesting some redundancy. The pair plot was skipped here since there are 8 numeric predictors, above the 6-feature threshold set in the pipeline.

Email Spam Dataset

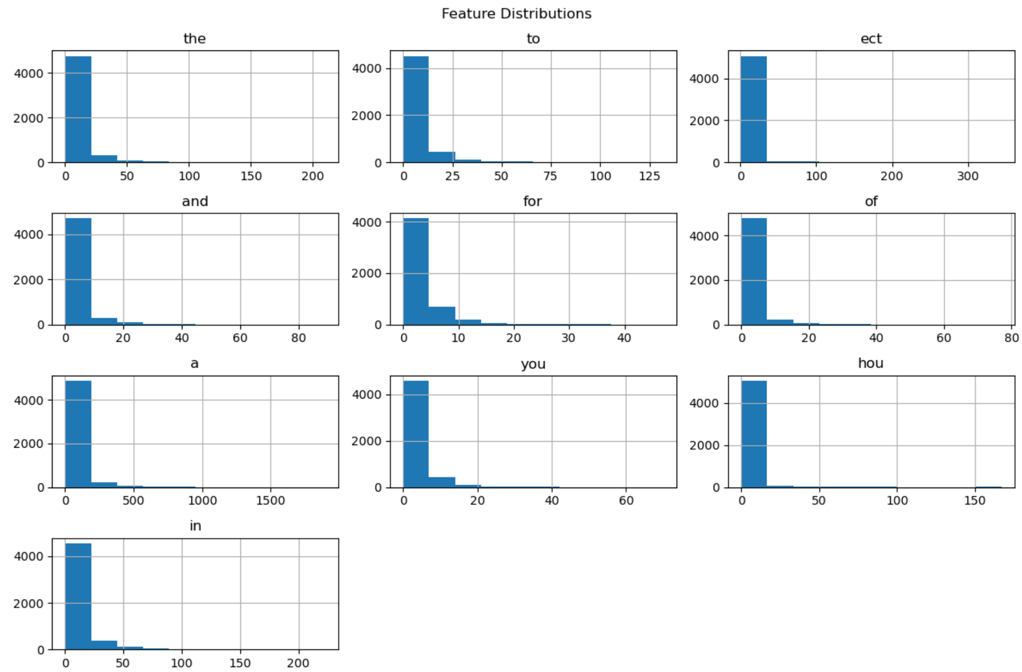

Figure 9: Feature distributions for a sample of word-frequency features (Email Spam dataset)

Inference: Almost all word-frequency features are extremely right-skewed, with the vast majority of emails having a value of zero for any given word and a small number of emails using that word heavily. This is typical of bag-of-words style features and suggests that raw counts may need log-transformation or that models robust to skewed, sparse features (such as Naive Bayes) are well suited to this data.

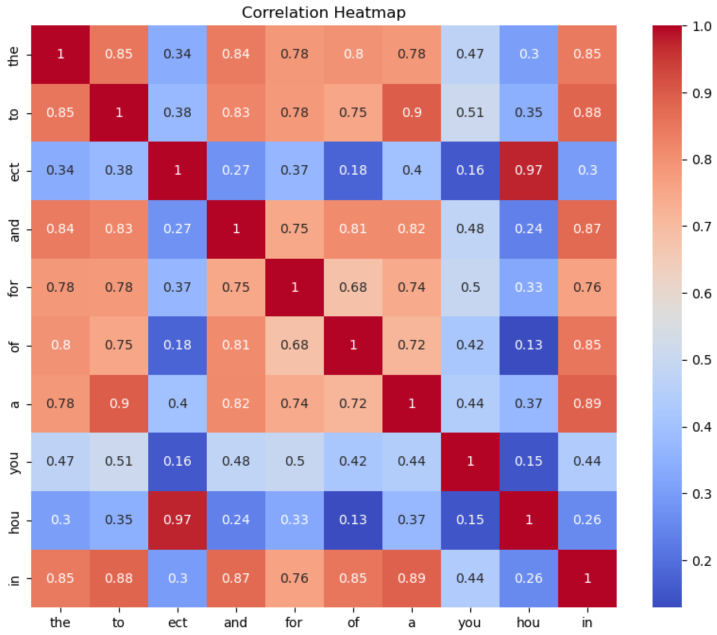

Figure 10: Correlation heatmap of a sample of Email Spam word-frequency features

Inference: Most word-frequency features are weakly correlated with each other, which is expected since each column represents a different, largely independent vocabulary term. The pair plot was skipped for this dataset since it has around 3,000 numeric columns, making a pairwise plot both uninformative and computationally impractical.

MNIST (Sampled) Dataset

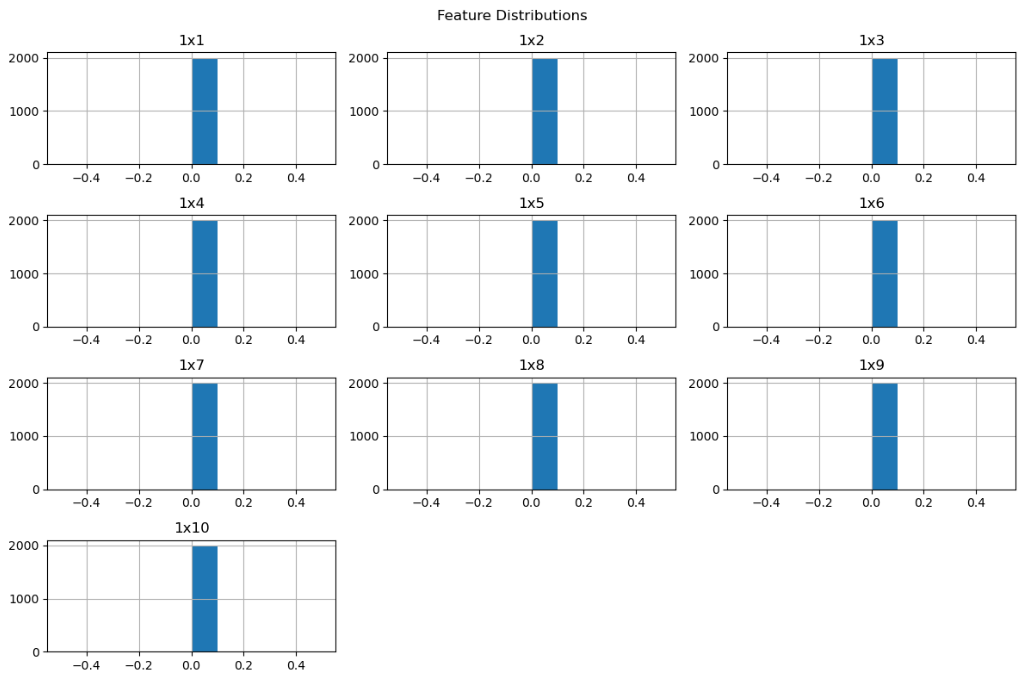

Figure 11: Feature distributions for a sample of pixel features (MNIST dataset)

Inference: Most pixel columns near the image borders (e.g. 1x1–1x9) are constant at zero across the sample, since handwritten digits are typically centred and rarely touch the image edges. This tells us that a large fraction of MNIST's 784 pixel features carry no information at all and could be safely dropped in later feature-selection steps.

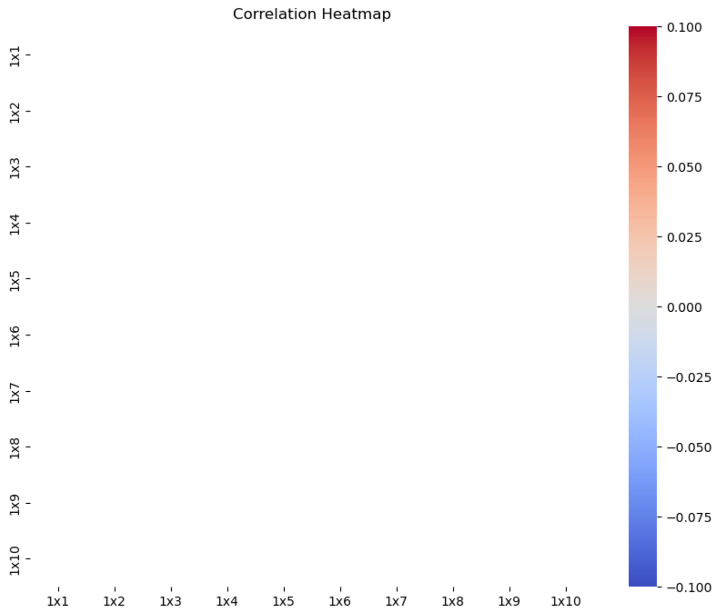

Figure 12: Correlation heatmap of a sample of MNIST pixel features

Inference: The heatmap could not be computed for several of the sampled border pixels because they have zero variance (constant value), which produces `NaN` correlations, visible as blank cells and a runtime warning during plotting. This is itself a useful diagnostic: it flags dead/uninformative features that should be removed before training a model on this dataset. The pair plot was skipped since MNIST has 784 numeric features.

5 Data Preprocessing

A second reusable function, `preprocessing_process()`, was applied to each dataset after EDA. It performs the following steps in order:

- **Task-type detection:** A helper function `detect_task_type()` inspects the target column's data type and number of unique values. If the target is of type `object/category`, or numeric with at most 20 unique values, the task is classified as **Classification**; otherwise it is treated as **Regression**. This automatically identified Iris, Diabetes, Email Spam, and MNIST as classification problems, and the Loan dataset (predicting `LoanAmount`) as a regression problem.
- **Missing value handling:** For the Diabetes dataset, biologically invalid zeros in Glucose, BloodPressure, SkinThickness, Insulin, and BMI were first converted to `NaN`. For all datasets, remaining numeric `NaN`s were filled with the column median (e.g. 14 values in Loan_Amount_Term and 50 in Credit_History for the Loan dataset; 5, 35, 227, 374, and 11 values respectively for Glucose, BloodPressure, SkinThickness, Insulin, and BMI in the Diabetes dataset). Rows still containing missing values after this step were dropped (79 rows removed from the Loan dataset).

- **Outlier handling:** For every numeric feature, the IQR method was used: $\text{IQR} = Q3 - Q1$, with values outside $[Q1 - 1.5 \times \text{IQR},\ Q3 + 1.5 \times \text{IQR}]$ clipped to the nearest bound rather than removed. This preserved sample size while limiting the influence of extreme values (for example, Insulin in the Diabetes dataset had 346 clipped outliers, reflecting its highly skewed distribution).

- **Feature scaling:** All numeric features (excluding the target) were standardized using `StandardScaler`, transforming each feature to zero mean and unit variance.

- **Train-test split:** Not performed at this stage; the pipeline returns a fully cleaned and scaled DataFrame together with the detected task type, ready to be split in a subsequent modelling step.

6 Machine Learning Algorithm

This experiment does not train a predictive model; it focuses on the supporting NumPy/SciPy/Pandas/M
Learn utilities that precede modelling. The key techniques used are:

- **Working principle:** `pandas` and `numpy` are used for vectorized data loading, summary statistics, and numeric operations; `matplotlib` and `seaborn` for visualization; and scikit-learn's `StandardScaler` for feature standardization, given by

$$z = \frac{x - \mu}{\sigma}$$

  where μ and σ are the mean and standard deviation of a feature computed on the data.

- **Mathematical formulation:** Outlier capping uses Tukey's IQR rule, $\text{IQR} = Q_3 - Q_1$, with bounds $[Q_1 - 1.5\,\text{IQR},\ Q_3 + 1.5\,\text{IQR}]$; values outside this range are clipped to the nearest bound.

- **Advantages:** A single generic pipeline can be reused across structurally very different datasets (few features vs. thousands, classification vs. regression targets) with minimal changes, which reduces code duplication and enforces a consistent preprocessing standard.

- **Limitations:** The automatic task-type detection rule (unique-value threshold of 20) is a heuristic and can misclassify targets in edge cases (e.g. a genuinely continuous target with few observed unique values). Median imputation and IQR clipping are simple, robust defaults but can mask important structure in features with legitimately skewed or multi-modal distributions, and the fixed pairplot/plot-column limits (6 features, first 10 columns) mean that some information is not visualized for high-dimensional datasets like Email Spam and MNIST.

7 Experimental Setup

Python Version	
Scikit-Learn Version	
IDE	Jupyter Notebook
Operating System	Windows
Hardware	

8 Hyperparameter Selection

Parameter	Value
Outlier threshold (IQR multiplier)	1.5
Task-type unique-value threshold	20
Pairplot feature limit	6 numeric features
Plotted columns limit (histograms)	First 10 numeric features

9 Results

This experiment produces cleaned and scaled data rather than predictive metrics; the table below summarizes the outcome of the pipeline on each dataset instead of classification/regression scores.

Dataset	Task Detected	Rows After Cleaning	Missing Values Handled
Iris	Classification	150	0
Loan Prediction	Regression	535	79 rows dropped, 64 values imputed
Diabetes	Classification	768	652 zero-as-missing values imputed
Email Spam	Classification	5,172	0
MNIST (sample)	Classification	2,000	0

10 Result Visualization

The dataset-wise figures in Section 5 (histograms, correlation heatmaps, and pair plots) constitute the result visualizations for this experiment, since no model was trained. Together they show: (i) which features are skewed and need transformation, (ii) which feature pairs are redundant or highly correlated, and (iii) how well classes separate visually before any modelling is attempted.

11 Discussion

The experiment confirmed that a single, generically-written EDA and preprocessing pipeline can be applied across datasets that differ enormously in scale – from 6 columns (Iris) to over 3,000 (Email Spam) – with only the target-column argument changing. The automatic task-type detector correctly separated the four classification datasets from the one regression dataset. The main limitation observed was with very high-dimensional datasets (Email Spam, MNIST): pairwise correlation and pairplot visualizations become impractical, and features with zero variance (e.g. constant-zero border pixels in MNIST) caused `NaN` values in the correlation matrix. A useful improvement would be to add automatic zero-variance feature dropping and, for text/pixel-style data, dimensionality reduction (e.g. PCA) before visualization.

12 Conclusion

This experiment demonstrated practical, hands-on use of NumPy, SciPy, Pandas, Matplotlib, Seaborn, and Scikit-Learn to build a reusable EDA and preprocessing pipeline. Applying the same pipeline to five structurally different datasets (Iris, Loan Prediction, Diabetes, Email Spam, and MNIST) highlighted both the strength of writing generic, parameterized preprocessing code and its limitations when applied to very high-dimensional or sparse data.

13 References

1. F. Pedregosa *et al.*, "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

2. C. R. Harris *et al.*, "Array programming with NumPy," *Nature*, vol. 585, pp. 357–362, 2020.

3. J. D. Hunter, "Matplotlib: A 2D Graphics Environment," *Computing in Science & Engineering*, vol. 9, no. 3, pp. 90–95, 2007.

4. W. McKinney, "Data Structures for Statistical Computing in Python," in *Proc. 9th Python in Science Conf.*, 2010, pp. 56–61.

5. UCI Machine Learning Repository, "Iris Data Set" and "Pima Indians Diabetes Data Set," [Online]. Available: https://archive.ics.uci.edu/.